PS#13 - Taylor series - Answer key

1. (AC 8a.3 Activity 8.3.4) Consider the function $f(x) = \dfrac{1}{1-x}$. Let's find the degree n Taylor polynomial approximation centered at $a = 0$ for f.

   (a) To begin finding $T_n(x)$, do the usual work of computing the various derivatives of f and their respective values at $a = 0$. Record the first five derivatives of $f(x)$ in the table below.

$$\begin{aligned} f(x) &= (1-x)^{-1} \\ f'(x) &= (-1)(1-x)^{-2}(-1) \\ f''(x) &= (-1)(-2)(1-x)^{-3}(-1)^2 \\ f'''(x) &= (-1)(-2)(-3)(1-x)^{-4}(-1)^3 \\ f^{(4)}(x) &= (-1)(-2)(-3)(-4)(1-x)^{-5}(-1)^4 \\ f^{(5)}(x) &= (-1)(-2)(-3)(-4)(-5)(1-x)^{-6}(-1)^5 \end{aligned}$$

   (b) Next, evaluate the derivatives you determined in (a) at $a = 0$ and use these to find the values of the coefficients of the Taylor polynomial centered at $a = 0$.

$$\begin{aligned} f(0) &= (1-0)^{-1} = 1 & c_0 &= f(0) = 1 \\ f'(0) &= (-1)(1-0)^{-2}(-1) & c_1 &= \frac{f'(0)}{1!} = \frac{1}{1!} = 1 \\ f''(0) &= (-1)(-2)(1-0)^{-3}(-1)^2 = 2 & c_2 &= \frac{f''(0)}{2!} = \frac{2}{2!} = 1 \\ f'''(0) &= (-1)(-2)(-3)(1-0)^{-4}(-1)^3 = 6 & c_3 &= \frac{f'''(0)}{3!} = \frac{6}{3!} = 1 \\ f^{(4)}(0) &= (-1)(-2)(-3)(-4)(1-0)^{-5}(-1)^4 = 24 & c_4 &= \frac{f^{(4)}(0)}{4!} = \frac{24}{4!} = 1 \\ f^{(5)}(0) &= (-1)(-2)(-3)(-4)(-5)(1-0)^{-6}(-1)^5 = 120 & c_5 &= \frac{f^{(5)}(0)}{5!} = \frac{120}{5!} = 1 \end{aligned}$$

   (c) What pattern do you observe in the value of c_k? State the degree 5 Taylor polynomial, $T_5(x)$, as well as the formula you expect for the general degree n Taylor polynomial, $T_n(x)$.

   Looks like c_k is always 1. This makes sense because the value of $f^{(k)}(0)$ is always $k!$ based on how the negatives and powers work out.

   So, $T_5(x) = 1 + x + x^2 + x^3 + x^4 + x^5$, and $T_n(x) = 1 + x + x^2 + \ldots + x^n$.

   (d) By identifying the value of r, explain why the degree n Taylor polynomial $T_n(x)$ can be thought of as a finite geometric series.

   The value of r is the ratio of the next term to the previous:

$$r = \frac{c_{n+1}x^{n+1}}{c_n x^n} = \frac{1 \cdot x^{n+1}}{1 \cdot x^n} = x.$$

   At any particular point x, this is indeed a fixed ratio – it doesn't depend at all on n.

(e) What is the Taylor series centered at $a = 0$ for $f(x) = \dfrac{1}{1-x}$?

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \ldots = \sum_{k=0}^{\infty} x^k.$$

2. (AC Activity 8a.4.2) Consider the infinite Taylor series given by

$$\begin{aligned} T(x) &= \sum_{k=1}^{\infty} \frac{1}{k \cdot 2^k}(x-1)^k \\ &= \frac{1}{1\cdot 2}(x-1) + \frac{1}{2\cdot 4}(x-1)^2 + \frac{1}{3\cdot 8}(x-1)^3 + \cdots + \frac{1}{n\cdot 2^n}(x-1)^n + \cdots. \end{aligned}$$

(a) As described in the statment of the Ratio Test, let $r_n(x)$ be the ratio of the $(n+1)$st term of $T(x)$ to the nth term of $T(x)$. Find the simplest formula that you can for $r_n(x)$.

$$\begin{aligned} r_n(x) &= \frac{a_{n+1}}{a_n} = \frac{a_{n+1}}{1} \cdot \frac{1}{a_n} \\ &= \frac{1\cdot(x-1)^{(n+1)}}{(n+1)\cdot 2^{(n+1)}} \cdot \frac{n\cdot 2^n}{1\cdot(x-1)^n} \\ &= \frac{n}{n+1} \cdot \frac{2^n}{2^{n+1}} \cdot \frac{(x-1)^{n+1}}{(x-1)^n} \\ &= \frac{n}{n+1} \cdot \frac{1}{2} \cdot \frac{x-1}{1}. \end{aligned}$$

(b) Let $r(x) = \lim_{n\to\infty} r_n(x)$. Evaluate this limit to find the simplest formula you can for $r(x)$.

$$\begin{aligned} r(x) &= \lim_{n\to\infty} r_n(x) = \lim_{n\to\infty} \left[\frac{1}{2} \cdot \frac{n}{n+1}(x-1)\right] \\ &= \frac{1}{2}(x-1) \lim_{n\to\infty} \frac{n}{n+1} = \frac{1}{2}(x-1)\cdot 1. \end{aligned}$$

(c) For what values of x is $|r(x)| < 1$? What does this tell us about $T(x)$ for these x-values?

$|r(x)| < 1$ when $\frac{1}{2}|x-1| < 1$, so when $|x-1| < 2$. This means "the distance between x and 1 is below 2," so $-1 < x < 3$. This is the "interval of convergence" – the x-values for which the ratio test guarantees this series will converge.

(d) Let $T_{10}(x)$ be the sum of the first 10 terms of $T(x)$, and let $f(x) = \ln(2) - \ln(3-x)$. Plot $f(x)$ and $T_{10}(x)$ on the same coordinate axes. What do you notice? What does this suggest about the series $T(x)$?

3. (AC 8a.4 Exercise 5) As we will see in more detail in the next section, we can use known Taylor series to obtain other Taylor series, and we preview that idea in this exercise.

(a) Calculate the first four derivatives of $\sin(x^2)$ and hence find the fourth order Taylor polynomial for $\sin(x^2)$ centered at $a = 0$.

$$\begin{aligned}
f(x) &= \sin(x^2) \\
f'(x) &= 2x\cos(x^2) \\
f''(x) &= 2x(-\sin(x^2)\cdot 2x) + 2\cos(x^2) \\
&= -4x^2\sin(x^2) + 2\cos(x^2) \\
f'''(x) &= -8x\sin(x^2) - 4x^2\cos(x^2)\cdot 2x + 2(-\sin(x^2)\cdot 2x) \\
&= -8x^3\cos(x^2) - 12x\sin(x^2) \\
f^{(4)}(x) &= -24x^2\cos(x^2) - 8x^3\cdot(-\sin(x^2)\cdot 2x) - 12\sin(x^2) - 12x(\cos(x^2)\cdot 2x) \\
&= -48x^2\cos(x^2) + (16x^4 - 12)\sin(x^2)
\end{aligned}$$

At $a = 0$:

$$\begin{aligned}
f(0) &= \sin(0) = 0 \\
f'(0) &= \cos(0)\cdot 2\cdot 0 = 0 \\
f''(0) &= -4\cdot 0^2\sin(0) + 2\cos(0) = 2 \\
f'''(0) &= -8\cdot 0^3\cos(0) - 12\cdot 0\sin(0) = 0 \\
f^{(4)}(0) &= -48\cdot 0^2\cos(0) + (16\cdot 0^4 - 12)\sin(0) = 0
\end{aligned}$$

So since a bunch of these derivatives are zero, $T_4(x) = 0 + 0x + \frac{2}{2!}x^2 + 0x^3 + 0x^4$ – that is, $T_4(x)$ is just x^2. (Boring!)

(b) Part (a) demonstrates the direct approach to finding Taylor polynomials and series. Next we utilize a known Taylor series to make the process simpler. Recall that the Taylor series centered at 0 for $f(x) = \sin(x)$ is

$$T(x) = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}. \tag{1}$$

i. Substitute x^2 for x in the Taylor series $T(x)$ given above. Write out the first several terms and compare to your work in part (a). Explain why the substitution in this problem should result in the Taylor series for $\sin(x^2)$ centered at 0.

$$\begin{aligned}
\text{Since } \sin(x) = T(x) &= \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}, \\
\text{it makes sense that } \sin(x^2) = T(x^2) &= \sum_{k=0}^{\infty} (-1)^k \frac{(x^2)^{2k+1}}{(2k+1)!} \\
&= \sum_{k=0}^{\infty} (-1)^k \frac{x^{4k+2}}{(2k+1)!}
\end{aligned}$$

The first few terms are:

$$T(x^2) = \underbrace{\frac{x^2}{1!}}_{k=0} - \underbrace{\frac{x^6}{3!}}_{k=1} + \underbrace{\frac{x^{10}}{5!}}_{k=2} - \underbrace{\frac{x^{14}}{7!}}_{k=3} - \dots$$

Makes sense that we didn't see anything past x^2 when we looked at just the first four derivatives.

ii. For what interval of x-values should we expect the Taylor series for $\sin(x^2)$ to converge?
Since we came up with this series by substituting x^2 into the Taylor series for $\sin(x)$, which converges for all real values of x, I think this series should converge for all real values of x also.

4. (Bonus task!) In this problem we're going to explore the Taylor series for $f(x) = \sqrt{x}$ centered at $x = 1$.

(a) Find the Taylor polynomials for $f(x) = \sqrt{x}$ up through order 5, centered at $x = 1$.

$$\begin{aligned}
f(x) &= x^{1/2} & f(1) &= 1 \\
f'(x) &= \frac{1}{2}x^{-1/2} & f'(1) &= \frac{1}{2} \\
f''(x) &= \frac{1}{2}\left(-\frac{1}{2}\right)x^{-3/2} & f''(1) &= \frac{1}{2}\left(-\frac{1}{2}\right) \\
f'''(x) &= \frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)x^{-5/2} & f'''(1) &= \frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right) \\
f^{(4)}(x) &= \frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)x^{-7/2} & f^{(4)}(1) &= \frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right) \\
f^{(5)}(x) &= \frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)\left(-\frac{7}{2}\right)x^{-9/2} & f^{(5)}(1) &= \frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)\left(-\frac{7}{2}\right)
\end{aligned}$$

Okay, so $T_5(x) = 1 + \frac{1}{2}(x-1) - \frac{1}{2!\cdot 2^2}(x-1)^2 + \frac{1\cdot 3}{3!\cdot 2^3}(x-1)^3 - \frac{1\cdot 3\cdot 5}{4!\cdot 2^4}(x-1)^4 + \frac{1\cdot 3\cdot 5\cdot 7}{5!\cdot 2^5}(x-1)^5$.

(b) Spot the pattern in the coefficients.
In the numerator we have an alternator $(-1)^{k+1}$ and the product of odd numbers. (If you want to read about "double factorials" that may be fun!) In the denominator we have a factorial and a power of 2:

$$c_k = (-1)^{k+1}\frac{1\cdot 3\cdot \ldots \cdot (2k-3)}{k!\cdot 2^k}.$$

(c) Use the pattern you found to do the ratio test.

$$|r_n(x)| = \left|\frac{c_{n+1}(x-1)^{n+1}}{c_n(x-1)^n}\right|$$

$$= \left| \frac{1 \cdot 3 \cdot \ldots \cdot (2n-3)(2n-1)(x-1)^{n+1}}{(n+1)! \cdot 2^{n+1}} \cdot \frac{n! \cdot 2^n}{1 \cdot 3 \cdot \ldots \cdot (2n-3)(x-1)^n} \right|$$

$$= \frac{(2n-1)}{(n+1) \cdot 2} \cdot |x-1|$$

$$|r(x)| = |x-1| \lim_{n \to \infty} \frac{2n-1}{2n+2} = |x-1| \cdot 1$$

(d) What is the "interval of convergence" – the x-values for which the Taylor series will converge? Why does your answer make sense based on the graph of the original function $f(x) = \sqrt{x}$?

This series will converge whenever $|r(x)| = |x-1| < 1$ – that is, for $0 < x < 2$. This makes sense because the graph of $\sqrt{x}$ stops at $x = 0$.

(e) Use $T_5(x)$ to find a good fraction approximation for $\sqrt{2} \approx 1.414\ldots$

$$T_5(2) = 1 + \frac{1}{2}(2-1) - \frac{1}{2! \cdot 2^2}(2-1)^2 + \frac{1 \cdot 3}{3! \cdot 2^3}(2-1)^3 - \frac{1 \cdot 3 \cdot 5}{4! \cdot 2^4}(2-1)^4 + \frac{1 \cdot 3 \cdot 5 \cdot 7}{5! \cdot 2^5}(2-1)^5$$

$$= 1 + \frac{1}{2} - \frac{1}{2! \cdot 2^2} + \frac{1 \cdot 3}{3! \cdot 2^3} - \frac{1 \cdot 3 \cdot 5}{4! \cdot 2^4} + \frac{1 \cdot 3 \cdot 5 \cdot 7}{5! \cdot 2^5}$$

$$= \frac{365}{256} = 1.42578125.$$

Not bad!